B2: Symmetry and Relativity — Model Solutions, Trinity Term 2022

Signature $(-,+,+,+)$; $x^\mu = (ct, x, y, z)$, $x_\mu = (-ct, x, y, z)$; $\partial_\mu \equiv \partial/\partial x^\mu$, $\partial^\mu \equiv \partial/\partial x_\mu$.

Question 1 — Lorentz covariance of Maxwell's equations

Problem. Identify ∂^μ vs ∂_μ among the operators in (3); show $\partial_\mu\partial^\mu$ is invariant; derive charge conservation and the 4-current; in Lorenz gauge show $\partial^\nu\partial_\nu A^a = -\mu_0 J^a$ and prove A^μ is a 4-vector; for a plane wave show one can choose $\vec{k}\cdot\vec{\epsilon} = 0$.

(a)

Apply the chain rule to a scalar Φ:

$$\frac{\partial\Phi}{\partial x'^\mu} = \frac{\partial x^\nu}{\partial x'^\mu}\frac{\partial\Phi}{\partial x^\nu}.$$

Read off the transformation of $\partial/\partial x^\mu$:

$$\partial'_\mu = \frac{\partial x^\nu}{\partial x'^\mu}\,\partial_\nu.$$

Compare with covariant rule (2): $\partial/\partial x^\mu$ transforms like x_μ:

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu}.$$

Identify the lower-index operator from (3) using $x^0 = ct$:

$$\boxed{\partial_\mu = \Big(\tfrac{1}{c}\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\Big).}$$

Raise with $\eta^{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$:

$$\partial^\mu = \eta^{\mu\nu}\partial_\nu.$$

Flip sign of time component only:

$$\boxed{\partial^\mu = \Big(-\tfrac{1}{c}\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\Big).}$$

(b)

Transform ∂'_μ as a covariant operator:

$$\partial'_\mu = \frac{\partial x^\alpha}{\partial x'^\mu}\,\partial_\alpha.$$

Transform ∂'^μ as a contravariant operator:

$$\partial'^\mu = \frac{\partial x'^\mu}{\partial x^\beta}\,\partial^\beta.$$

Multiply the two:

$$\partial'_\mu \partial'^\mu = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x'^\mu}{\partial x^\beta}\,\partial_\alpha \partial^\beta.$$

Use the inverse–matrix identity from the hint:

$$\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x'^\mu}{\partial x^\beta} = \delta^\alpha_\beta.$$

Contract on β:

$$\boxed{\partial'_\mu \partial'^\mu = \partial_\alpha \partial^\alpha.}$$

(c)

Start from Ampère–Maxwell:

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t}.$$

Take divergence; use $\nabla \cdot (\nabla \times \vec{B}) = 0$:

$$0 = \mu_0 \nabla \cdot \vec{J} + \frac{1}{c^2}\frac{\partial}{\partial t}(\nabla \cdot \vec{E}).$$

Substitute Gauss $\nabla \cdot \vec{E} = \rho/\epsilon_0$:

$$0 = \mu_0 \nabla \cdot \vec{J} + \frac{1}{c^2 \epsilon_0}\frac{\partial \rho}{\partial t}.$$

Apply $c^2 = 1/(\mu_0\epsilon_0)$ so $1/(c^2\epsilon_0) = \mu_0$:

$$0 = \mu_0 \nabla \cdot \vec{J} + \mu_0 \frac{\partial \rho}{\partial t}.$$

Divide by μ_0:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0.$$

Expand $\partial_\mu J^\mu = (1/c)\partial_t J^0 + \nabla \cdot \vec{J}$ and match:

$$\boxed{J^\mu = (c\rho,\ \vec{J}), \qquad \partial_\mu J^\mu = 0.}$$

∂_μ is covariant by (a); quotient rule then makes J^μ a 4-vector.

(d)

Insert $\vec{E} = -\nabla\phi - \partial_t \vec{A}$ into Gauss $\nabla \cdot \vec{E} = \rho/\epsilon_0$:

$$-\nabla^2 \phi - \frac{\partial}{\partial t}(\nabla \cdot \vec{A}) = \rho/\epsilon_0.$$

Substitute $\vec{B} = \nabla \times \vec{A}$ into Ampère–Maxwell:

$$\nabla \times (\nabla \times \vec{A}) = \mu_0 \vec{J} + \frac{1}{c^2}\frac{\partial}{\partial t}\Big(-\nabla\phi - \frac{\partial \vec{A}}{\partial t}\Big).$$

Expand using $\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$:

$$-\nabla^2 \vec{A} + \nabla(\nabla \cdot \vec{A}) + \frac{1}{c^2}\nabla\frac{\partial \phi}{\partial t} + \frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} = \mu_0 \vec{J}.$$

Write the Lorenz condition with $A^\mu = (\phi/c, \vec{A})$ explicitly:

$$\frac{1}{c^2}\frac{\partial \phi}{\partial t} + \nabla \cdot \vec{A} = 0.$$

Substitute $\nabla \cdot \vec{A} = -(1/c^2)\partial_t \phi$ into the Gauss equation:

$$-\nabla^2 \phi - \frac{\partial}{\partial t}\Big[-\frac{1}{c^2}\frac{\partial \phi}{\partial t}\Big] = \rho/\epsilon_0.$$

Simplify the time term:

$$-\nabla^2 \phi + \frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} = \rho/\epsilon_0.$$

Apply the same condition to the Ampère equation, noting $\nabla(\nabla \cdot \vec{A}) = -(1/c^2)\nabla \partial_t \phi$:

$$-\nabla^2 \vec{A} - \frac{1}{c^2}\nabla \frac{\partial \phi}{\partial t} + \frac{1}{c^2}\nabla \frac{\partial \phi}{\partial t} + \frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} = \mu_0 \vec{J}.$$

Cancel the gradient terms:

$$-\nabla^2 \vec{A} + \frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} = \mu_0 \vec{J}.$$

Apply $\partial^\nu \partial_\nu = -(1/c^2)\partial_t^2 + \nabla^2$ to the scalar equation:

$$-\partial^\nu \partial_\nu \phi = \rho/\epsilon_0.$$

Divide by c on both sides:

$$-\partial^\nu \partial_\nu (\phi/c) = \rho/(c\epsilon_0).$$

Substitute $A^0 = \phi/c$ on the left and $1/(c\epsilon_0) = c\mu_0$ on the right:

$$-\partial^\nu \partial_\nu A^0 = c\mu_0 \rho.$$

Use $J^0 = c\rho$:

$$\partial^\nu \partial_\nu A^0 = -\mu_0 J^0.$$

Apply the same operator to the vector equation:

$$-\partial^\nu \partial_\nu \vec{A} = \mu_0 \vec{J}.$$

Identify $A^i = A_i$ (spatial) and $J^i = J_i$:

$$\partial^\nu \partial_\nu A^i = -\mu_0 J^i.$$

Combine into a single 4-equation:

$$\boxed{\partial^\nu \partial_\nu A^a = -\mu_0 J^a.}$$

RHS is a 4-vector by (c); $\partial^\nu \partial_\nu$ is a scalar by (b); quotient rule: A^a is a 4-vector.

(e)

Differentiate the plane wave $A^b = \epsilon^b \sin(k^\mu x_\mu)$ using $\partial_b(k^\mu x_\mu) = k_b$:

$$\partial_b A^b = \epsilon^b k_b \cos(k \cdot x).$$

Impose Lorenz gauge $\partial_b A^b = 0$ for all x:

$$\epsilon^b k_b = 0.$$

Apply $\partial^\nu \partial_\nu A^b = 0$ (vacuum wave equation from (d)):

$$-\epsilon^b k^\nu k_\nu \sin(k \cdot x) = 0.$$

Read off the vacuum dispersion relation:

$$k^\nu k_\nu = 0.$$

Apply a residual gauge transformation $A^\mu \to A^\mu + \partial^\mu \chi$, taking $\chi = \chi_0 \cos(k \cdot x)$:

$$\partial^\mu \chi = -\chi_0 k^\mu \sin(k \cdot x).$$

Compute $\Box \chi = \partial^\mu \partial_\mu \chi$:

$$\Box \chi = -\chi_0 (k^\mu k_\mu) \cos(k \cdot x) = 0,$$

so the gauge transformation preserves the Lorenz condition. Read off the shift of the polarisation:

$$\epsilon^b \longrightarrow \epsilon^b - \chi_0 k^b.$$

Choose $\chi_0 = \epsilon^0 / k^0$ to kill the time component:

$$\epsilon^0 \longrightarrow \epsilon^0 - (\epsilon^0 / k^0) k^0 = 0.$$

Split Lorenz $\epsilon^b k_b = 0$ with new $\epsilon^0 = 0$:

$$0 = \epsilon^0 k_0 + \epsilon^i k_i = \epsilon^i k_i.$$

Spatial indices have $\epsilon^i = \epsilon_i$ and $k^i = k_i$, hence $\epsilon^i k_i = \vec{\epsilon} \cdot \vec{k}$:

$$\boxed{\vec{k} \cdot \vec{\epsilon} = 0.}$$

Question 2 — Pion decay kinematics

Problem. ($c = 1$.) Find the CM-frame velocity for N particles; show $\sin^2(A/2) = M_\pi^2/(4E_1E_2)$ for $\pi^0 \to \gamma\gamma$; with isotropic CM emission derive $\mathrm{d}N/\mathrm{d}A$; find the minimum and peak of the distribution.

(a)

Sum 4-momenta in the lab frame:

$$P^\mu = \Big(\sum_i E_i,\ \sum_i \vec{p}_i \Big).$$

The CM frame is defined by vanishing total 3-momentum:

$$\vec{P}'_{\text{tot}} = \sum_i \vec{p}'_i = 0.$$

Boost the spatial part along $\vec{\beta}$ (sign convention of eq. (1) of the paper):

$$\vec{P}'_{\rm tot} = \gamma\big(\vec{P}_{\rm tot} - \vec{\beta}E_{\rm tot}\big).$$

The question's convention defines $\vec{\beta}$ as the velocity of the lab frame relative to the CM frame, i.e. the boost from CM to lab has parameter $\vec{\beta}$. Equivalently, to transform from lab to CM frame we apply a boost with parameter $-\vec{\beta}$:

$$\vec{P}'_{\rm tot} = \gamma\big(\vec{P}_{\rm tot} + \vec{\beta}E_{\rm tot}\big).$$

Set $\vec{P}'_{\rm tot} = 0$ and solve for $\vec{\beta}$:

$$\boxed{\vec{\beta} = -\frac{\vec{P}_{\rm tot}}{E_{\rm tot}} = -\frac{\sum_i \vec{p}_i}{\sum_i E_i}.}$$

(b)

Apply 4-momentum conservation:

$$p_\pi = p_1 + p_2.$$

Square both sides; use $p_\pi^2 = -M_\pi^2$ in signature $(-,+,+,+)$:

$$-M_\pi^2 = p_1^2 + p_2^2 + 2\,p_1\!\cdot\!p_2.$$

Use $p_i^2 = 0$ for massless photons:

$$-M_\pi^2 = 2\,p_1\!\cdot\!p_2.$$

Expand $p_1\!\cdot\!p_2 = -E_1E_2 + \vec{p}_1\!\cdot\!\vec{p}_2$:

$$M_\pi^2 = 2(E_1E_2 - \vec{p}_1\!\cdot\!\vec{p}_2).$$

Insert $|\vec{p}_i| = E_i$ and $\vec{p}_1\!\cdot\!\vec{p}_2 = E_1E_2\cos A$:

$$M_\pi^2 = 2E_1E_2(1-\cos A).$$

Apply the half-angle identity $1-\cos A = 2\sin^2(A/2)$:

$$M_\pi^2 = 4E_1E_2\sin^2(A/2).$$

Solve for $\sin^2(A/2)$:

$$\boxed{\sin^2(A/2) = \frac{M_\pi^2}{4E_1E_2}.} \tag{1}$$

(c)

In the CM frame the photons are back-to-back with $E_1^* = E_2^* = M_\pi/2$. Boost photon 1 (angle θ_* to the boost direction) to the lab:

$$E_1 = \gamma\tfrac{M_\pi}{2}(1+\beta\cos\theta_*).$$

Boost photon 2 (angle $\pi-\theta_*$):

$$E_2 = \gamma\tfrac{M_\pi}{2}(1-\beta\cos\theta_*).$$

Multiply the two energies:

$$E_1E_2 = \gamma^2\frac{M_\pi^2}{4}(1-\beta^2\cos^2\theta_*).$$

Insert into (1):

$$\sin^2(A/2) = \frac{1}{\gamma^2(1-\beta^2\cos^2\theta_*)}. \tag{2}$$

Invert for $1-\beta^2\cos^2\theta_*$:

$$1-\beta^2\cos^2\theta_* = \frac{1}{\gamma^2\sin^2(A/2)}.$$

Solve for $\cos^2\theta_*$ and use $\gamma^2\beta^2 = \gamma^2 - 1$:

$$\cos^2\theta_* = \frac{\gamma^2\sin^2(A/2)-1}{(\gamma^2-1)\sin^2(A/2)}. \tag{3}$$

Differentiate (2) with respect to θ_*:

$$2\sin(A/2)\cos(A/2)\cdot\tfrac{1}{2}\,\frac{\mathrm{d}A}{\mathrm{d}\theta_*} = -\frac{1}{\gamma^2(1-\beta^2\cos^2\theta_*)^2}\cdot\frac{\mathrm{d}}{\mathrm{d}\theta_*}(1-\beta^2\cos^2\theta_*).$$

Evaluate $\mathrm{d}(1-\beta^2\cos^2\theta_*)/\mathrm{d}\theta_* = 2\beta^2\sin\theta_*\cos\theta_*$:

$$\sin(A/2)\cos(A/2)\,\frac{\mathrm{d}A}{\mathrm{d}\theta_*} = -\frac{2\beta^2\sin\theta_*\cos\theta_*}{\gamma^2(1-\beta^2\cos^2\theta_*)^2}.$$

Take absolute values (count rates are positive):

$$\sin(A/2)\cos(A/2)\,|\mathrm{d}A| = \frac{2\beta^2\sin\theta_*\cos\theta_*}{\gamma^2(1-\beta^2\cos^2\theta_*)^2}\,|\mathrm{d}\theta_*|.$$

Substitute $\gamma^2(1-\beta^2\cos^2\theta_*) = 1/\sin^2(A/2)$:

$$\sin(A/2)\cos(A/2)\,|\mathrm{d}A| = 2\beta^2\gamma^2\sin\theta_*\cos\theta_*\sin^4(A/2)\,|\mathrm{d}\theta_*|.$$

Solve for $\mathrm{d}\theta_*/\mathrm{d}A$:

$$\frac{\mathrm{d}\theta_*}{\mathrm{d}A} = \frac{\cos(A/2)}{2\beta^2\gamma^2\sin^3(A/2)\sin\theta_*\cos\theta_*}.$$

Apply the CM rule $\mathrm{d}N/\mathrm{d}\theta_* = a\sin\theta_*$:

$$\frac{\mathrm{d}N}{\mathrm{d}A} = a\sin\theta_*\cdot\frac{\mathrm{d}\theta_*}{\mathrm{d}A} = \frac{a\cos(A/2)}{2\beta^2\gamma^2\sin^3(A/2)\cos\theta_*}.$$

From (3), with $\gamma^2 - 1 = \beta^2\gamma^2$:

$$\cos\theta_* = \frac{\sqrt{\gamma^2\sin^2(A/2)-1}}{\beta\gamma\sin(A/2)}.$$

Substitute and simplify the denominator:

$$\beta^2\gamma^2\sin^3(A/2)\cos\theta_* = \beta\gamma\sin^2(A/2)\sqrt{\gamma^2\sin^2(A/2)-1}.$$

Insert into the count rate:

$$\frac{\mathrm{d}N}{\mathrm{d}A} = \frac{a\cos(A/2)}{2\beta\gamma\sin^2(A/2)\sqrt{\gamma^2\sin^2(A/2)-1}}.$$

Read off $f(\gamma) = 1/(2\gamma)$:

$$\boxed{\frac{\mathrm{d}N}{\mathrm{d}A} = \frac{a}{2\gamma}\,\frac{\cos(A/2)}{\beta\sin^2(A/2)\sqrt{\gamma^2\sin^2(A/2)-1}}, \quad f(\gamma) = \frac{1}{2\gamma}.}$$

(d)

From (2), $\sin^2(A/2)$ is minimised when $1 - \beta^2\cos^2\theta_*$ is maximised. Set $\cos^2\theta_* = 0$ (photons perpendicular to boost):

$$\sin^2(A_{\min}/2) = \frac{1}{\gamma^2}.$$

Take the positive root and double:

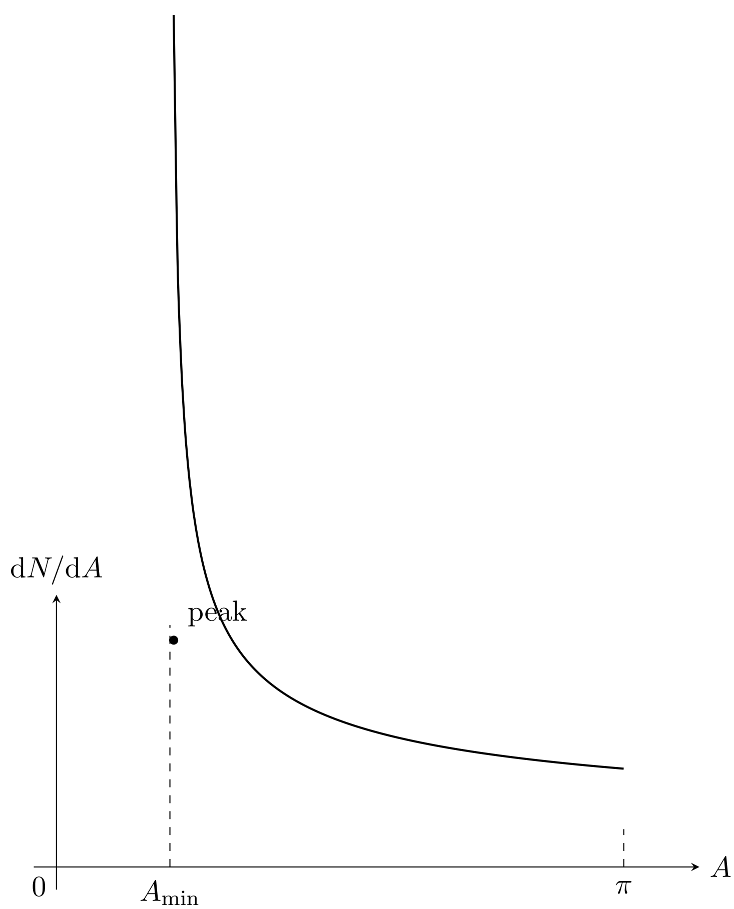
$$\boxed{A_{\min} = 2\arcsin(1/\gamma).}$$

Inspect the count rate near $A_{\min}$: the factor $\sqrt{\gamma^2\sin^2(A/2) - 1}$ vanishes there, so $1/\sqrt{\cdots}$ diverges:

$$\lim_{A\to A_{\min}^+} \frac{\mathrm{d}N}{\mathrm{d}A} \to \infty.$$

The other factors $\cos(A/2)$, $\sin^2(A/2)$, $1/\beta$ are finite and smooth at $A_{\min}$. Hence the peak coincides with the minimum:

$$\boxed{A_{\text{peak}} = A_{\min} = 2\arcsin(1/\gamma).}$$



Question 3 — 4-velocity, addition, two-star problem

Problem. (a) Show $\mathrm{d}\tau = \mathrm{d}t/\gamma_u$ and $u^\alpha = (c\gamma_u, \gamma_u\vec{u})$. (b) Find $\vec{w}$ in standard-configuration B. (c) For arbitrary boost $\vec{v}$, derive $|\vec{w}|$. (d) Stars S_1, S_2 at $c/2$ in A; light from S_2 at ω_0 received by S_1 at ω_0: find the propagation angle in S_1's frame, and the propagation direction in A.

(a)

Equate the invariant interval in frame A and in the rest frame (where $\mathrm{d}\vec{x} = 0$):

$$-c^2\mathrm{d}\tau^2 = -c^2\mathrm{d}t^2 + \mathrm{d}\vec{x} \cdot \mathrm{d}\vec{x}.$$

Substitute $\mathrm{d}\vec{x} = \vec{u}\,\mathrm{d}t$:

$$-c^2\mathrm{d}\tau^2 = -c^2\mathrm{d}t^2 + u^2\mathrm{d}t^2.$$

Divide by $-c^2$:

$$\mathrm{d}\tau^2 = \mathrm{d}t^2(1 - u^2/c^2).$$

Take the positive root and apply $\gamma_u = (1 - u^2/c^2)^{-1/2}$:

$$\boxed{\mathrm{d}\tau = \mathrm{d}t/\gamma_u.}$$

Differentiate $x^\alpha = (ct, \vec{x})$ with respect to τ:

$$u^\alpha = \frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau} = \frac{\mathrm{d}t}{\mathrm{d}\tau}\frac{\mathrm{d}x^\alpha}{\mathrm{d}t}.$$

Substitute $\mathrm{d}t/\mathrm{d}\tau = \gamma_u$ and $\mathrm{d}x^\alpha/\mathrm{d}t = (c, \vec{u})$:

$$\boxed{u^\alpha = (c\gamma_u,\ \gamma_u\vec{u}).}$$

(b)

Apply the standard boost ($\beta_v = v/c$) to each component of $u^\alpha = (c\gamma_u, \gamma_u\vec{u})$:

$$u'^0 = \gamma_v(u^0 - \beta_v u^1), \quad u'^1 = \gamma_v(u^1 - \beta_v u^0), \quad u'^2 = u^2, \quad u'^3 = u^3.$$

Substitute the 4-velocity components into u'^0:

$$u'^0 = \gamma_v\gamma_u c(1 - vu_x/c^2).$$

Substitute into u'^1:

$$u'^1 = \gamma_v\gamma_u(u_x - v).$$

Form $w_x = cu'^1/u'^0$:

$$w_x = \frac{u_x - v}{1 - vu_x/c^2}.$$

Form $w_y = cu'^2/u'^0 = c\gamma_u u_y/u'^0$:

$$w_y = \frac{u_y}{\gamma_v(1 - vu_x/c^2)}.$$

Form w_z identically:

$$\boxed{w_x = \frac{u_x - v}{1 - vu_x/c^2}, \quad w_y = \frac{u_y}{\gamma_v(1 - vu_x/c^2)}, \quad w_z = \frac{u_z}{\gamma_v(1 - vu_x/c^2)}.}$$

(c)

Decompose along $\hat{v}$:

$$\vec{u}_{\parallel} = (\vec{u}\cdot\hat{v})\hat{v}, \qquad \vec{u}_{\perp} = \vec{u} - \vec{u}_{\parallel}.$$

Generalise (b) by sending $u_x \to \vec{u}_{\parallel}$, $u_{y,z} \to \vec{u}_{\perp}$:

$$\vec{w} = \frac{1}{1 - \vec{v}\cdot\vec{u}/c^2}\Big[(\vec{u}_{\parallel} - \vec{v}) + \vec{u}_{\perp}/\gamma_v\Big].$$

Square using $(\vec{u}_{\parallel} - \vec{v}) \perp \vec{u}_{\perp}$:

$$|\vec{w}|^2 = \frac{|\vec{u}_{\parallel} - \vec{v}|^2 + |\vec{u}_{\perp}|^2/\gamma_v^2}{(1 - \vec{v}\cdot\vec{u}/c^2)^2}.$$

Substitute $1/\gamma_v^2 = 1 - v^2/c^2$ into the numerator:

$$|\vec{u}_{\parallel} - \vec{v}|^2 + |\vec{u}_{\perp}|^2/\gamma_v^2 = |\vec{u}_{\parallel} - \vec{v}|^2 + |\vec{u}_{\perp}|^2 - (v^2/c^2)|\vec{u}_{\perp}|^2.$$

Combine the first two terms via Pythagoras (orthogonal decomposition of $\vec{u} - \vec{v}$):

$$|\vec{u}_{\parallel} - \vec{v}|^2 + |\vec{u}_{\perp}|^2 = |\vec{u} - \vec{v}|^2 = (\vec{v} - \vec{u})\cdot(\vec{v} - \vec{u}).$$

Use $\vec{u}_{\parallel} \times \vec{v} = 0$ so $\vec{u}\times\vec{v} = \vec{u}_{\perp}\times\vec{v}$, with $|\vec{u}_{\perp}\times\vec{v}| = |\vec{u}_{\perp}||\vec{v}|$:

$$v^2|\vec{u}_{\perp}|^2 = |\vec{v}\times\vec{u}|^2 = (\vec{v}\times\vec{u})\cdot(\vec{v}\times\vec{u}).$$

Assemble the numerator:

$$|\vec{u}_{\parallel} - \vec{v}|^2 + |\vec{u}_{\perp}|^2/\gamma_v^2 = (\vec{v} - \vec{u})\cdot(\vec{v} - \vec{u}) - \frac{1}{c^2}(\vec{v}\times\vec{u})\cdot(\vec{v}\times\vec{u}).$$

Take the positive square root:

$$\boxed{|\vec{w}| = \frac{\sqrt{(\vec{v} - \vec{u})\cdot(\vec{v} - \vec{u}) - (1/c^2)(\vec{v}\times\vec{u})\cdot(\vec{v}\times\vec{u})}}{1 - \vec{v}\cdot\vec{u}/c^2}.} \tag{4}$$

(d)

Set $\vec{v} = \vec{u}_1 = (c/2, 0, 0)$ and $\vec{u} = \vec{u}_2 = (c/(2\sqrt{2}), c/(2\sqrt{2}), 0)$. Compute the dot product:

$$\vec{v}\cdot\vec{u} = \frac{c}{2}\cdot\frac{c}{2\sqrt{2}} = \frac{c^2}{4\sqrt{2}}.$$

Compute $|\vec{v} - \vec{u}|^2 = v^2 + u^2 - 2\vec{v}\cdot\vec{u}$ with $v^2 = u^2 = c^2/4$:

$$|\vec{v} - \vec{u}|^2 = \frac{c^2}{4} + \frac{c^2}{4} - \frac{c^2}{2\sqrt{2}} = \frac{c^2}{2}\left(1 - \frac{1}{\sqrt{2}}\right).$$

Use the hint $|\vec{v}\times\vec{u}|^2 = v^2u^2 - (\vec{v}\cdot\vec{u})^2$:

$$|\vec{v}\times\vec{u}|^2 = \frac{c^4}{16} - \frac{c^4}{32} = \frac{c^4}{32}.$$

Insert into (4):

$$V_{\text{rel}}^2 = \frac{(c^2/2)(1 - 1/\sqrt{2}) - c^2/32}{(1 - 1/(4\sqrt{2}))^2}.$$

Plug in $1/\sqrt{2} = 0.7071$, $1/(4\sqrt{2}) = 0.1768$ in the numerator:

$$\mathrm{Num}/c^2 = 0.5 \times 0.2929 - 0.03125 = 0.11520.$$

Square the denominator:

$$\mathrm{Den} = (1 - 0.17678)^2 = (0.82322)^2 = 0.67770.$$

Divide:

$$V_{\mathrm{rel}}^2/c^2 = 0.11520/0.67770 = 0.17000.$$

Take square root:

$$V_{\mathrm{rel}}/c = 0.41231.$$

Compute the Lorentz factor:

$$\Gamma = \frac{1}{\sqrt{1 - 0.17000}} = \frac{1}{\sqrt{0.83000}} = 1.09764.$$

Apply the relativistic Doppler formula (source S_2, receiver S_1, $\hat{n}$ light direction in S_1):

$$\omega_{\mathrm{obs}} = \frac{\omega_0}{\Gamma(1 - \vec{V}_{\mathrm{rel}} \cdot \hat{n}/c)}.$$

Set $\omega_{\mathrm{obs}} = \omega_0$:

$$\Gamma(1 - \vec{V}_{\mathrm{rel}} \cdot \hat{n}/c) = 1.$$

Solve for $\cos\alpha = \hat{V}_{\mathrm{rel}} \cdot \hat{n}$:

$$\cos\alpha = \frac{c}{V_{\mathrm{rel}}}\left(1 - \frac{1}{\Gamma}\right).$$

Numerical: compute $1 - 1/\Gamma$:

$$1 - 1/\Gamma = 1 - 0.91105 = 0.08895.$$

Divide by V_{rel}/c:

$$\cos\alpha = 0.08895/0.41231 = 0.21574.$$

Take arccos:

$$\boxed{\alpha = \arccos\big[(c/V_{\mathrm{rel}})(1 - 1/\Gamma)\big] \approx 77.55^\circ.}$$

Direction in frame A. Both stars move at the same speed $c/2$ in A, so the configuration is invariant under reflection about the angle bisector of $\hat{u}_1, \hat{u}_2$. A Doppler-equality light ray maps to itself under this reflection, hence lies on the bisector:

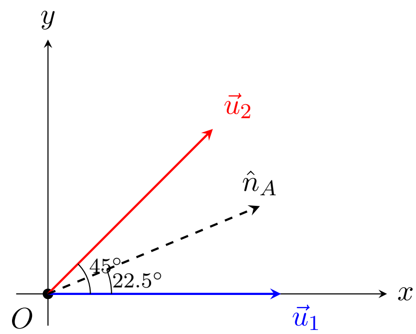
$$\hat{n}_A \propto \hat{u}_1 + \hat{u}_2.$$

Substitute $\hat{u}_1 = (1, 0, 0)$ and $\hat{u}_2 = (1/\sqrt{2}, 1/\sqrt{2}, 0)$:

$$\boxed{\hat{n}_A \propto \left(1 + \tfrac{1}{\sqrt{2}},\ \tfrac{1}{\sqrt{2}},\ 0\right).}$$

Compute the angle to the x-axis (half of the 45° between $\hat{u}_1, \hat{u}_2$):

$$\theta_A = \arctan\left[\frac{1/\sqrt{2}}{1 + 1/\sqrt{2}}\right] = 22.5^\circ.$$



Question 4 — Constant force, 4-force, synchrotron

Problem. (a) For constant $\vec{F}$ from rest find $\vec{v}(t)$ and show $|\vec{v}| \to c$. (b) Show $F^\alpha = (\gamma \vec{F} \cdot \vec{v}/c, \gamma \vec{F})$ with $\vec{F} = m\gamma^3[\vec{a} + \vec{v} \times (\vec{v} \times \vec{a})/c^2]$. (c) Compare radiative loss in linear vs circular accelerators. (d) Choose particle/machine to minimise loss.

(a)

Integrate $\vec{F} = \mathrm{d}(\gamma m \vec{v})/\mathrm{d}t$ from rest, with constant $\vec{F}$:

$$\gamma m \vec{v} = \vec{F} t.$$

Take magnitude squared:

$$\gamma^2 m^2 v^2 = F^2 t^2.$$

Substitute $\gamma^2 v^2 = v^2/(1 - v^2/c^2)$:

$$\frac{m^2 v^2}{1 - v^2/c^2} = F^2 t^2.$$

Clear the denominator:

$$m^2 v^2 = F^2 t^2 - F^2 t^2 v^2/c^2.$$

Collect v^2 terms:

$$v^2 \big(m^2 + F^2 t^2/c^2\big) = F^2 t^2.$$

Solve for v^2:

$$v^2 = \frac{F^2 t^2}{m^2 + F^2 t^2/c^2} = \frac{(Ft/m)^2}{1 + (Ft/(mc))^2}.$$

Take the positive root and align direction with $\vec{F}$:

$$\boxed{\vec{v}(t) = \frac{\vec{F} t/m}{\sqrt{1 + (Ft/(mc))^2}}.}$$

Take $t \to \infty$; numerator $\sim Ft/m$, denominator $\sim Ft/(mc)$:

$$|\vec{v}(t)| \to \frac{Ft/m}{Ft/(mc)} = c.$$

Bounded limit:

$$\boxed{\lim_{t\to\infty} |\vec{v}| = c.}$$

(b)

Use $p^0 = m\gamma c = E/c$ and $\mathrm{d}\tau = \mathrm{d}t/\gamma$ in the time component:

$$F^0 = \frac{\mathrm{d}p^0}{\mathrm{d}\tau} = \gamma\frac{\mathrm{d}p^0}{\mathrm{d}t}.$$

Substitute $p^0 = E/c$:

$$F^0 = \frac{\gamma}{c}\frac{\mathrm{d}E}{\mathrm{d}t}.$$

Apply work-energy $\mathrm{d}E/\mathrm{d}t = \vec{F}\cdot\vec{v}$:

$$F^0 = \gamma\vec{F}\cdot\vec{v}/c.$$

Apply $\mathrm{d}\tau = \mathrm{d}t/\gamma$ to the spatial component, with $\vec{F} = \mathrm{d}\vec{p}/\mathrm{d}t$:

$$F^i = \frac{\mathrm{d}p^i}{\mathrm{d}\tau} = \gamma\frac{\mathrm{d}p^i}{\mathrm{d}t} = \gamma F^i.$$

Combine into a 4-vector:

$$\boxed{F^\alpha = (\gamma\vec{F}\cdot\vec{v}/c,\ \gamma\vec{F}).}$$

Expand $\vec{F} = \mathrm{d}(m\gamma\vec{v})/\mathrm{d}t$:

$$\vec{F} = m\frac{\mathrm{d}\gamma}{\mathrm{d}t}\vec{v} + m\gamma\vec{a}.$$

Substitute $\mathrm{d}\gamma/\mathrm{d}t = \gamma^3\vec{v}\cdot\vec{a}/c^2$:

$$\vec{F} = \frac{m\gamma^3}{c^2}(\vec{v}\cdot\vec{a})\vec{v} + m\gamma\vec{a}.$$

Apply BAC–CAB with $\vec{A} = \vec{B} = \vec{v}, \vec{C} = \vec{a}$:

$$\vec{v}\times(\vec{v}\times\vec{a}) = (\vec{v}\cdot\vec{a})\vec{v} - v^2\vec{a}.$$

Rearrange for $(\vec{v}\cdot\vec{a})\vec{v}$:

$$(\vec{v}\cdot\vec{a})\vec{v} = \vec{v}\times(\vec{v}\times\vec{a}) + v^2\vec{a}.$$

Substitute into $\vec{F}$:

$$\vec{F} = \frac{m\gamma^3}{c^2}\left[\vec{v}\times(\vec{v}\times\vec{a}) + v^2\vec{a}\right] + m\gamma\vec{a}.$$

Group the $\vec{a}$ terms:

$$\vec{F} = m\gamma\left(1 + \frac{\gamma^2 v^2}{c^2}\right)\vec{a} + \frac{m\gamma^3}{c^2}\vec{v}\times(\vec{v}\times\vec{a}).$$

Apply $1 + \gamma^2 v^2/c^2 = \gamma^2$:

$$\vec{F} = m\gamma^3\vec{a} + \frac{m\gamma^3}{c^2}\vec{v}\times(\vec{v}\times\vec{a}).$$

Factor:

$$\boxed{\vec{F} = m\gamma^3\left[\vec{a} + \frac{\vec{v}\times(\vec{v}\times\vec{a})}{c^2}\right].}$$

(c)

Linear case ($\vec{a} \parallel \vec{v}$, so $\vec{v} \times \vec{a} = 0$). From (b):

$$F = m\gamma^3 a.$$

Evaluate the Minkowski norm of $F^\alpha = (\gamma F v/c, \gamma F \hat{v})$ in $(-,+,+,+)$:

$$F^\alpha F_\alpha = -\gamma^2 F^2 v^2/c^2 + \gamma^2 F^2 = \gamma^2 F^2 (1 - v^2/c^2) = F^2.$$

Substitute $F^2 = m^2\gamma^6 a^2$:

$$\left| \frac{\mathrm{d}p^\alpha}{\mathrm{d}\tau} \frac{\mathrm{d}p_\alpha}{\mathrm{d}\tau} \right| = m^2\gamma^6 a^2.$$

Plug into the magnitude of Larmor's formula (taking $P > 0$ as physical):

$$P_{\text{lin}} = \frac{q^2}{6\pi\epsilon_0 m^2 c^3}\, m^2\gamma^6 a^2 = \frac{q^2\gamma^6 a^2}{6\pi\epsilon_0 c^3}.$$

Circular case ($\vec{a} \perp \vec{v}$). Apply BAC–CAB:

$$\vec{v} \times (\vec{v} \times \vec{a}) = (\vec{v} \cdot \vec{a})\vec{v} - v^2\vec{a} = -v^2\vec{a}.$$

Substitute into the (b) expression:

$$\vec{F} = m\gamma^3 \left[\vec{a} - \frac{v^2}{c^2}\vec{a} \right].$$

Factor and use $\gamma^2(1 - v^2/c^2) = 1$:

$$\vec{F} = m\gamma^3 (1 - v^2/c^2)\vec{a} = m\gamma\vec{a}.$$

With $\vec{F} \cdot \vec{v} = 0$, the time component F^0 vanishes; the norm becomes:

$$\left| \frac{\mathrm{d}p^\alpha}{\mathrm{d}\tau} \frac{\mathrm{d}p_\alpha}{\mathrm{d}\tau} \right| = \gamma^2 F^2 = \gamma^2 (m\gamma a)^2 = m^2\gamma^4 a^2.$$

Plug into the magnitude of Larmor's formula:

$$P_{\text{circ}} = \frac{q^2}{6\pi\epsilon_0 m^2 c^3}\, m^2\gamma^4 a^2 = \frac{q^2\gamma^4 a^2}{6\pi\epsilon_0 c^3}.$$

Take the ratio at the same γ and a (as the question requests):

$$\frac{P_{\text{circ}}}{P_{\text{lin}}} = \frac{\gamma^4 a^2}{\gamma^6 a^2}.$$

Simplify:

$$\boxed{\frac{P_{\text{circ}}}{P_{\text{lin}}} = \frac{1}{\gamma^2}, \qquad \text{equivalently } \frac{P_{\text{lin}}}{P_{\text{circ}}} = \gamma^2.}$$

(d)

At fixed final energy E, write $\gamma = E/(mc^2)$. For a ring of fixed radius R, the centripetal acceleration is:

$$a = \frac{v^2}{R} \to \frac{c^2}{R} \quad (\gamma \gg 1).$$

Substitute into $P_{\rm circ}$:

$$P_{\rm circ} = \frac{q^2\gamma^4 a^2}{6\pi\epsilon_0 c^3} \propto \frac{\gamma^4}{R^2} \propto \frac{E^4}{m^4 R^2}.$$

In a linear machine, $\vec{a} \parallel \vec{v}$ and the radiated power for fixed accelerating force F uses $F = m\gamma^3 a$:

$$P_{\rm lin} = \frac{q^2\gamma^6 a^2}{6\pi\epsilon_0 c^3} = \frac{q^2 F^2}{6\pi\epsilon_0 m^2 c^3}.$$

At fixed F, $P_{\rm lin} \propto 1/m^2$ and is independent of γ, whereas $P_{\rm circ}$ grows as $\gamma^4 \propto E^4/m^4$. Compare mass ratio $m_\mu/m_e \approx 207$:

$$(m_\mu/m_e)^2 \approx 4.3 \times 10^4, \quad (m_\mu/m_e)^4 \approx 1.8 \times 10^9.$$

Both factors favour the muon, and the linear geometry removes the γ^4 enhancement. Choose:

$$\boxed{\text{muon in a linear accelerator.}}$$

Reasons: (i) linear geometry replaces $a = c^2/R$ with the (small) accelerating gradient, eliminating the γ^4 ring enhancement; (ii) the muon's larger rest mass suppresses Larmor by an extra $1/m^2$ at fixed accelerating force.